

Experiment 8: Travelling Salesman Problem (TSP)

Aim

To implement the Travelling Salesman Problem using Python.

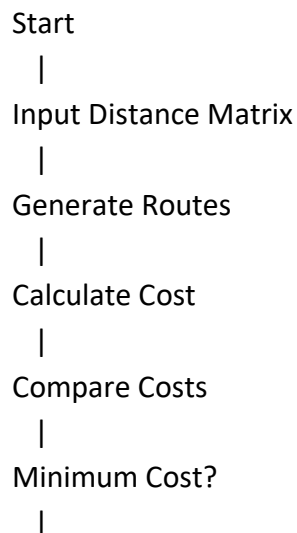
Objective

- To determine the minimum cost path.
- To visit every city exactly once and return to the starting city.

Algorithm

1. Store distances between cities.
2. Generate all possible routes.
3. Calculate the total cost for each route.
4. Select the route with minimum cost.
5. Display the optimal route.

Flowchart



Display Best Route

|
End

Program

```
from itertools import permutations

graph = [
    [0,10,15,20],
    [10,0,35,25],
    [15,35,0,30],
    [20,25,30,0]
]

cities = [1,2,3]
min_cost = float('inf')

for path in permutations(cities):
    cost = graph[0][path[0]]
    for i in range(len(path)-1):
        cost += graph[path[i]][path[i+1]]
    cost += graph[path[-1]][0]

    if cost < min_cost:
        min_cost = cost
        best = path

print("Minimum Cost:", min_cost)
print("Route: 0 ->", *best, "-> 0")
```

Output

Minimum Cost: 80
Route: 0 -> 1 3 2 -> 0

Result

The minimum travelling cost was obtained successfully.

Conclusion

The Travelling Salesman algorithm finds the shortest possible tour covering all cities.